Reg. No: …….

Name: …….

MAHATHMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025

THIRD SEMESTER

Discipline Specific Core Course (DSC A) MG3DSCFSC201- FORENSIC CHEMISTRY

FORENSIC SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours
Maximum Mark: 50

Remember (K), Understanding(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skills(S), Interest(I) and Appreciation (AP)

Students should attempt at least one question from each course outcome to enhance their overall outcomes attainability

Part A

Short Answer Type Questions
Answer any **10** questions
Each question carries 2 marks
Short Answer Questions (2 Mark)

1. Define forensic chemistry? *[U] / [CO3]*
2. What is the job profile of a forensic chemist?
3. What is the purpose of phenolphthalein in trap cases? *[U] / [CO3]*
4. Define inflammables. Give one example?*[U] / [CO5]*
5. What is the principle behind UV-Visible spectrophotometry?
6. What is a presumptive test?*[U] / [CO3]*
7. What are trace elements?*[U] / [CO4]*
8. Define denatured spirit.*[U] / [CO2]*
9. What is the purpose of phenolphthalein in trap cases?*[U] / [CO3]*
10. What is the function of a binder paint in paint composition?*[U] / [CO2]*

11. Mention the colour tests to detect ethanol and methanol in alcoholic beverages.*[U] / [CO3]*
12. Draw and explain the fire tetrahedron?*[U] / [CO4]*

Answer any **5** questions

Short Essay type question (4 Marks)

Each question carries 4 marks

1. Discuss the importance of gold and other chelating agents as crime exhibits?*[U] / [CO4]*
2Describe the importance of analysing petrol, diesel and kerosene in a fire and arson crime scenes?*[U] / [CO3]*
3. What is the difference between solvent extraction and solid-phase extraction?*[U] / [CO2]*
4. Explain the Iodoform test for ethyl alcohol.*[A] / [CO2]*
5. What are the main components of paint? Explain each component. *[A] / [CO1]*
6. Explain the forensic significance of chemicals used in trap cases.*[U] / [CO4]*
7. Write a note on the role of elemental analysis in forensic exhibits.*[U] / [CO4]*

Answer any 1 question

Essay (10 Marks)

Each question carries 10 marks

1. What are forensic exhibits? Describe their forensic importance in crime scene investigation?*[U] / [CO4]*
2. Discuss the methods used for identification and estimation of alcohol in forensic samples, along with the relevant laws.*[A] / [CO3]*